

Recall:

1) The least square solutions of a matrix equation $A\mathbf{x} = \mathbf{b}$ are the solutions of the equation

$$A\mathbf{x} = \text{proj}_{\text{Col}(A)} \mathbf{b}$$

2) If $A\mathbf{x} = \mathbf{b}$ is a consistent equation, then $\mathbf{b} \in \text{Col}(A)$, and $\text{proj}_{\text{Col}(A)} \mathbf{b} = \mathbf{b}$. In such case the least square solutions of $A\mathbf{x} = \mathbf{b}$ are just the ordinary solutions.

3) If $A\mathbf{x} = \mathbf{b}$ is inconsistent, then the least square solutions are the best substitute for the (nonexistent) ordinary solutions.

4) If $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is an orthogonal basis of a subspace V of $\mathbb{R}^n$ then

$$\text{proj}_V \mathbf{w} = \left(\frac{\mathbf{w} \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \right) \mathbf{v}_1 + \dots + \left(\frac{\mathbf{w} \cdot \mathbf{v}_k}{\mathbf{v}_k \cdot \mathbf{v}_k} \right) \mathbf{v}_k$$

5) If $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is an arbitrary basis of V then we can use the Gram-Schmidt process to obtain an orthogonal basis of V .

How to compute least square solutions of $Ax = b$
(version 1.0)

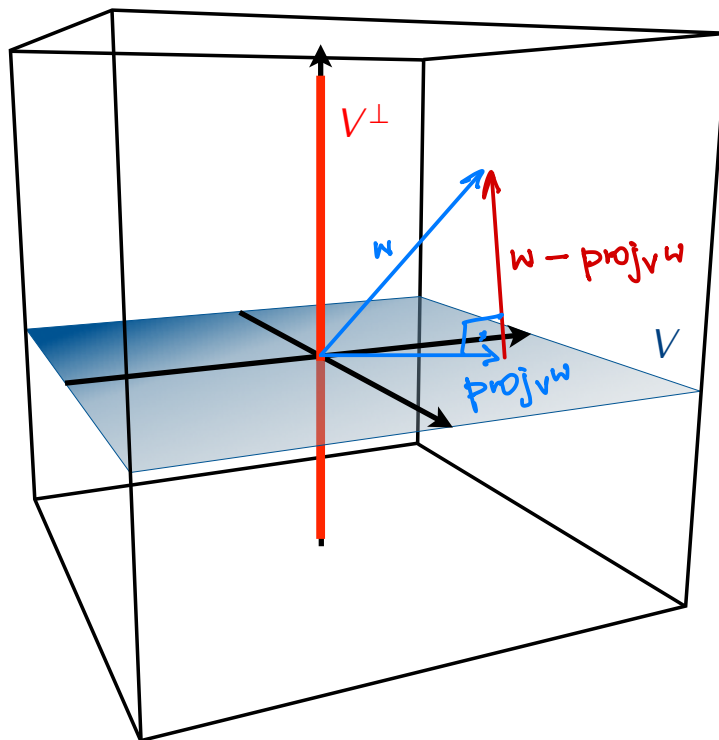
- 1) Compute a basis of $\text{Col}(A)$.
- 2) Use the Gram-Schmidt process to get an orthogonal basis of $\text{Col}(A)$.
- 3) Use the orthogonal basis to compute $\text{proj}_{\text{Col}(A)} \mathbf{b}$.
- 4) Compute solutions of the equation $Ax = \text{proj}_{\text{Col}(A)} \mathbf{b}$.

Next goal: Simplify this.

Definition

If V is a subspace of $\mathbb{R}^n$ then the *orthogonal complement* of V is the set $V^\perp$ of all vectors orthogonal to V :

$$V^\perp = \{\mathbf{w} \in \mathbb{R}^n \mid \mathbf{w} \cdot \mathbf{v} = 0 \text{ for all } \mathbf{v} \in V\}$$



Note: If $V \subseteq \mathbb{R}^n$, $\mathbf{w} \in \mathbb{R}^n$ then $(\mathbf{w} - \text{proj}_V \mathbf{w}) \in V^\perp$

Proposition

If V is a subspace of $\mathbb{R}^n$ then:

- 1) $V^\perp$ is also a subspace of $\mathbb{R}^n$.
- 2) For $\mathbf{w} \in \mathbb{R}^n$, the projection $\text{proj}_V \mathbf{w}$ is the unique vector such that $\text{proj}_V \mathbf{w} \in V$ and $\mathbf{w} - \text{proj}_V \mathbf{w} \in V^\perp$.

Definition

If A is an $m \times n$ matrix then the *row space* of A is the subspace $\text{Row}(A)$ of $\mathbb{R}^n$ spanned by rows of A .

Example

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$$\text{Row}(A) = \text{Span} \left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} \right)$$

Proposition

If A is a matrix then

$$\text{Row}(A)^\perp = \text{Nul}(A)$$

Note: If $A = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_m \end{bmatrix}$ rows of A

then:

$$Av = \begin{bmatrix} r_1 \cdot v \\ r_2 \cdot v \\ \vdots \\ r_m \cdot v \end{bmatrix}$$

↑
dot products

e.g.:

$$\begin{matrix} r_1 \\ r_2 \end{matrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \cdot \begin{matrix} v \\ 7 \\ 8 \\ 9 \end{matrix} = \begin{bmatrix} 1 \cdot 7 + 2 \cdot 8 + 3 \cdot 9 \\ 4 \cdot 7 + 5 \cdot 8 + 6 \cdot 9 \end{bmatrix} = \begin{bmatrix} r_1 \cdot v \\ r_2 \cdot v \end{bmatrix}$$

Proof of Proposition:

Let $A = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_m \end{bmatrix}$. We have: $v \in \text{Nul}(A)$

$$\Updownarrow \\ Av = 0$$

$$\Updownarrow \\ r_1 \cdot v = 0, r_2 \cdot v = 0, \dots, r_m \cdot v = 0$$

$$\Updownarrow \\ v \in \text{Row}(A)^\perp$$

Corollary

If A is a matrix then

$$\text{Col}(A)^\perp = \text{Nul}(A^T)$$

Note: $\text{Col}(A) = \text{Row}(A^T)$

e.g.:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

$$\text{Col}(A) = \text{Span} \left(\begin{bmatrix} 1 \\ 4 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \end{bmatrix} \right) \quad \text{Row}(A^T) = \text{Span} \left(\begin{bmatrix} 1 \\ 4 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \end{bmatrix} \right)$$

Proof of Corollary

$$\text{Col}(A)^\perp = \text{Row}(A^T)^\perp = \text{Nul}(A^T).$$

Back to least square solutions

Theorem

A vector $\hat{x}$ is a least square solution of a matrix equation

$$Ax = b$$

if and only if $\hat{x}$ is an ordinary solution of the equation

$$(A^T A)x = A^T b$$

Proof: If $\hat{x}$ is a least squares solution of $Ax=b$ then:

$$A\hat{x} = \text{proj}_{\text{Col}(A)} b$$

This gives:

$$(b - A\hat{x}) = (b - \text{proj}_{\text{Col}(A)} b) \in \text{Col}(A)^\perp = \text{Nul}(A^T)$$

We obtain:

$$A^T (b - A\hat{x}) = 0$$

$$A^T b - A^T A\hat{x} = 0$$

$$A^T b = A^T A\hat{x}$$

This shows that $\hat{x}$ is a solution of the equation

$$A^T A x = A^T b$$

The proof of the other implication is similar.

Definition

The equation

$$(A^T A)x = A^T b$$

is called the *normal equation* of $Ax = b$.

How to compute least square solutions of $Ax = b$
(version 2.0)

- 1) Compute $A^T A, A^T b$.
- 2) Solve the normal equation $(A^T A)x = A^T b$.

Example. Compute least square solutions of the following equation:

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}}_A \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}}_b$$

Note:

$$\begin{cases} x_1 + x_2 = 1 \\ 2x_2 = 2 \\ 0 = 3 \end{cases}$$

- no solutions !

Solution:

$$A^T = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 5 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$$

The normal equation:

$$A^T A x = A^T b$$
$$\begin{bmatrix} 1 & 1 \\ 1 & 5 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$$

augmented matrix:

$$\left[\begin{array}{cc|c} 1 & 1 & 1 \\ 1 & 5 & 5 \end{array} \right] \xrightarrow{\text{row reduction}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

least sq. solution : $\hat{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$